Sarthak Deore

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Summary

Computer Science undergraduate at **VIT–AP** (CGPA: **8.93**) with experience in **AI/ML, Computer Vision, and Full-Stack Development**. Built and deployed AI-powered applications spanning forensic video analysis, agentic systems, predictive modeling, and full-stack platforms.

Projects

TraceX – Multi-Vendor DVR/NVR Forensics Platform GitHub | Live | v1.0.1 *Aug 2026 – Oct 2026*

- Built a **vendor-agnostic forensics platform** (team of 5) that parses raw DVR/NVR disk images, recovers **deleted footage**, and normalizes recordings into a unified timeline with chain-of-custody and integrity hashing.
- Designed an extensible **parser architecture** (`BaseDVRParser`) with a Hikvision parser covering HIKBTREE filesystem parsing, unallocated-space carving, and MPEG-PS → MP4 conversion via FFmpeg.
- Developed an **AI engine** combining YOLO, Grounding DINO, and a CPU-only HOG/MOG2 detector with multi-frame tracking, event building, and collision/sudden-deceleration heuristics.
- Implemented a FastAPI + SQLAlchemy backend, Typer-based CLI/TUI, and a React + TypeScript dashboard with Supabase auth, forensic report generation, and an extensive pytest suite.

TripMind – Agentic AI Travel Planner GitHub *Jun 2026 – Jul 2026*

- Built an autonomous itinerary planner using a **LangChain + LangGraph ReAct agent** powered by Groq LLaMA 3.3 70B.
- Integrated tools for flight search, hotel recommendations, live weather (Open-Meteo), and budget estimation, with the agent reasoning across multi-step decisions.
- Generated structured day-wise itineraries with justified tool selection (`selection_reason`) for full explainability.

F1 Race Prediction & Visualization System GitHub | Model *Mar 2026 – Present*

- Developed Random Forest models using 3+ seasons of FastF1 telemetry achieving **94% podium accuracy** (AUC 0.93) and **87% winner accuracy**.
- Built and deployed a Streamlit application with telemetry visualization, live race replay, and a Deep Q-Network racing agent using PyTorch.

GraamGyaan – Rural Learning Platform GitHub | Live *Aug 2025 – Sep 2025*

- Developed a full-stack STEM learning platform using React, Node.js, Supabase, and Google Gemini API with **50+ learning modules**.
- Optimized mobile performance, reducing load time by approximately **30%**.

Education

Vellore Institute of Technology – AP University *2024 – Present*

B.Tech in Computer Science and Engineering **CGPA: 8.93**

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Database Systems, Operating Systems, Computer Networks

Achievement: Smart India Hackathon 2025 National Finalist (Rank 7)

Senior Secondary (12th) **84.17%**

Secondary (10th) **91%**

Skills

Languages: Python, Java, JavaScript, TypeScript, C, C++
Frameworks: FastAPI, React.js, Next.js, Node.js, Vite, Streamlit, Tailwind CSS, Supabase
Databases: PostgreSQL, MySQL, MongoDB, SQLAlchemy
AI/ML: Scikit-learn, PyTorch, OpenCV, YOLO, Grounding DINO, LangChain, LangGraph, Pandas, NumPy, NLP, Random Forest, Deep Q-Networks, Object Tracking
Tools: Git, Docker, FFmpeg, Pytest, Postman, Vercel, Figma
APIs: Google Gemini API, Groq API, FastF1 API, Open-Meteo API, REST APIs